

Least-cost cable route for a wind farm grid connection

Gronau-Epe, North Rhine-Westphalia · Cost-distance analysis in ArcGIS Pro (Spatial Analyst)

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The task, the cost parameters and the scenario data were set by the course. The data preparation, the analysis – including the comparison of all three connection points – the map and this assessment are my own work.

1 Question

A planned wind farm near Gronau-Epe has to be connected to one of three transformers on the nearest high-voltage line. Where does the cheapest cable route run, what does it cost, and which cadastral parcels does it cross – how many, and with what total area? Because the choice of connection point matters as much as the route itself, all three points were compared, not only the cheapest.

2 Data and source

Dataset	Content	Source
DTM	Digital terrain model, 1 m cells	Geobasis NRW, dl-de/zero-2-0
ALKIS	Buildings, roads, paths, watercourses, standing water, woodland, scrub, bog; 27,584 parcels	Geobasis NRW, dl-de/zero-2-0
Scenario	Wind farm transformer, 3 connection points, 19 contaminated sites	GIS-Akademie Berlin (exercise data)

The study area is the DTM extent, 4.6 × 4.5 km at the Dutch border. The DTM came in a custom projection equivalent to EPSG:4647 (zone number in the easting), the vector data in EPSG:25832; its cell size of 0.999889 m shows it had already been resampled before delivery. I reprojected it once to EPSG:25832 with bilinear resampling to the 2 m cells the task allows, on an even-metre grid, and used it as extent, snap raster and mask for every later step.

3 Method

The cost surface gives a cost in euros per metre and is additive:

- **Slope** (Surface Parameters, 3 × 3 window, degrees), reclassified: 0–3° → 3, 3–7° → 4, 7–15° → 7, 15–30° → 15, > 30° → 50.
- **Surcharges:** woodland and scrub +100, paths +500, roads +1,500, contaminated sites +2,000, watercourses +2,500.
- **Barriers:** buildings, standing water and bog, set to NoData (no route possible).

The layers were rasterised in one batch and combined in the Raster Calculator, with Con(IsNull()) so that "outside a polygon" counts as zero instead of spreading NoData. Distance Accumulation from the wind farm (geodesic) and Optimal Path As Line with path type "each cell" traced one route to every connection point, each with its path cost. Parcels crossed by the cheapest route were selected by intersection and summed with their official ALKIS area.

Checks: all points lie on valid cost cells and all rasters share one grid; a tool warning about zero costs (010779) was checked against the cost surface (minimum 3 €/m) and had no effect.

4 Result

Connection point	Route length	Model cost	Mean cost	Straight line
North	3.01 km	€61,942	€20.6/m	2.6 km
South	3.24 km	€83,450	€25.8/m	2.2 km
East	4.34 km	€91,603	€21.1/m	2.8 km

The cheapest route leads to the northern connection point: 3.0 km for €61,942. It crosses 28 parcels with an official area of 683,853 m² (68.4 ha); the geometric area is 91 m² larger (0.013 %), so the official value is the one reported.

Two findings stand out. The nearest point is not the cheapest: South is closest in a straight line, but its route costs 35 % more than North. And cost is driven by crossings, not length: 3 km at the base rate

would cost about €9,000, so roughly 85 % of the total comes from a few crossings. Slope hardly matters – 89 % of the area is below 3°, under 0.7 % steeper than 15°.

5 Analytical limits

- **Per-metre costs cannot tell crossing from following.** Crossing a road would in practice be one horizontal drilling at a one-off cost; the model charges it cell by cell.
- **The route looks realistic for the wrong reason.** For almost a kilometre it runs alongside the Amtsvennweg, but just outside the road parcel, on private land, because the parcel itself costs €1,500/m. Real cables go inside the public road parcel, in the verge, precisely because it has a single owner.
- **Parcels are an output, not a cost.** The task sums whole parcel areas (68 ha) for a cable a few metres wide; what matters for land rights – owners, metres per parcel, easement strip – is not in the model.
- **The connection point is not the planner's choice alone.** Under § 8 EEG the default is the suitable point with the shortest straight-line distance – here South – unless another is technically and economically more favourable, and the grid operator may assign a different one. The comparison informs that decision; it does not make it.
- **Slope at 2 m measures micro-relief** (ditch banks, embankments, DTM noise under vegetation), not the gradient relevant for trenching.
- **Not modelled:** protected areas, water protection zones, existing utilities, archaeology, soil, grid capacity. Costs are model values from the exercise parameters, not engineering estimates.